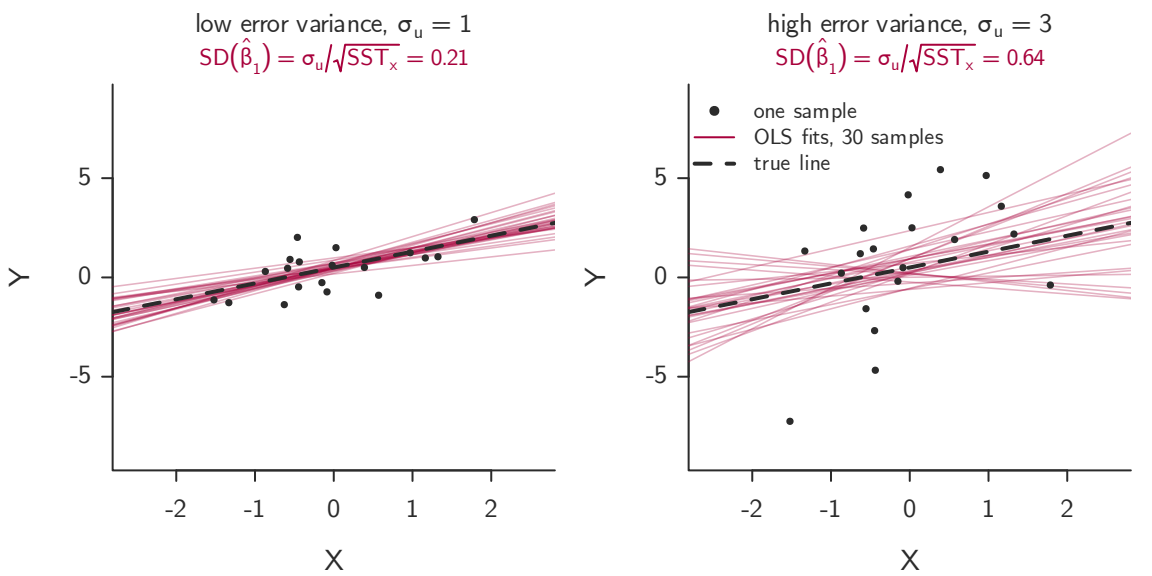
low error variance, $\sigma_u = 1$
$SD(\hat{\beta}_1) = \sigma_u / \sqrt{SST_x} = 0.21$
high error variance, $\sigma_u = 3$
$SD(\hat{\beta}_1) = \sigma_u / \sqrt{SST_x} = 0.64$
one sample
OLS fits, 30 samples
true line
Y
X
5
0
-5
-2
-1
0
1
2